

Enhanced Passive Radar DOA Estimation Using Wigner-Ville Distribution and MUSIC Algorithm in Noisy Environments

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Abstract—Accurate direction of arrival (DOA) estimation in noisy and interference-prone environments remains a critical challenge in modern radar systems, particularly when targets are closely spaced. This work introduces a Wigner-Ville Distribution (WVD) assisted MUSIC algorithm that utilizes the high-resolution time-frequency characteristics of WVD to enhance subspace separation and improve estimation accuracy. By incorporating WVD-based preprocessing, the proposed method significantly mitigates the impact of noise and electronic countermeasures, ensuring reliable detection of multiple targets with minimal angular separation. Extensive simulations using passive radar configurations demonstrate that the WVD-MUSIC approach achieves notable improvements in position and velocity root mean square errors compared to conventional MUSIC and short-time Fourier transform-based MUSIC techniques. The results affirm the robustness of the proposed method for tracking in dynamic and cluttered scenarios, making it a promising solution for defense and civilian applications requiring precise DOA estimation under challenging conditions.

Index Terms—Cramer-Rao Lower Bound, Direction of Arrival, Gaussian Noise, Wigner-Ville Distribution, MUSIC, Passive Radar

I. INTRODUCTION

Radar systems have long been the cornerstone of surveillance, navigation, and defense applications, providing critical information on the location and movement of targets. A fundamental task in radar signal processing is Direction of Arrival (DOA) estimation, which refers to determining the angle at which a signal arrives at an array of sensors. Accurate estimation of DOA is crucial for applications such as air traffic control, maritime navigation, electronic warfare, and emerging technologies such as autonomous vehicles. In

particular, the need to resolve targets with very small angular separations has become increasingly important due to the growing density of aerial and surface objects in operational environments [1].

Active radar systems traditionally have dominated the field by transmitting their own waveforms and analyzing the echoes reflected from the targets. However, these systems face two significant challenges: detectability and susceptibility to electronic countermeasures (ECM). To overcome these issues, passive radar systems have gained significant attention in recent years. Passive radars do not transmit signals; instead, they exploit existing non-cooperative illumination sources such as commercial radio, television, or cellular signals to track targets. This not only makes the radar system covert and less vulnerable to jamming but also offers economic and operational benefits by reducing hardware complexity. Despite these advantages, passive systems operate in interference-rich environments, and therefore demand advanced signal processing techniques for robust DOA estimation [2].

Over the years, several algorithms have been developed for DOA estimation, each with its own advantages and limitations. Early techniques relied on beamforming methods, which, while simple, suffer from limited resolution due to array aperture constraints. To overcome these limitations, subspace-based algorithms such as Multiple Signal Classification (MUSIC) and Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT) were introduced. MUSIC, in particular, has been widely adopted because of its ability to

separate signal and noise subspaces, leading to high-resolution estimation even in low signal-to-noise ratio (SNR) scenarios. However, these methods assume stationary signals and often degrade in performance when targets are closely spaced or in the presence of strong interference [3], [4].

In recent years, time-frequency analysis techniques have been integrated into DOA estimation frameworks to handle non-stationary signals. Methods based on the Short-Time Fourier Transform (STFT) and Stockwell Transform have been proposed to improve the resolution of signals with varying frequency content. While these techniques introduce adaptability through windowing, they are still constrained by trade-offs between time and frequency resolution. Moreover, their performance deteriorates under conditions of high noise, clutter, or electronic countermeasures, which are common in real-world radar deployments [5].

Despite significant progress, existing methods face three major challenges. First, resolving closely spaced targets remains difficult for algorithms relying on fixed-window time-frequency transforms, as they cannot adequately capture fine signal variations without sacrificing resolution. Second, in ECM-rich and noisy environments, covariance matrix estimation becomes unreliable, leading to incorrect subspace separation and poor DOA estimates. Third, most current approaches do not fully exploit the joint time-frequency characteristics of signals, which are essential for scenarios involving maneuvering targets and dynamic clutter [6], [7].

To address these limitations, this work introduces a Wigner-Ville Distribution (WVD)-assisted MUSIC algorithm. The WVD offers superior time-frequency localization compared to conventional transforms, making it highly effective for analyzing non-stationary signals with overlapping spectra [8]. By integrating WVD-based preprocessing with the MUSIC framework, the proposed approach enhances subspace orthogonality, leading to significant improvements in angular resolution and robustness against noise and interference. This methodology is particularly suited for passive radar applications, where reliance on non-cooperative sources and cluttered environments makes DOA estimation a challenging task [9].

The key contributions of the work include the proposal of a novel DOA estimation algorithm that combines the high-resolution time-frequency analysis of WVD with the subspace decomposition capabilities of MUSIC. Utilize WVD preprocessing to improve covariance matrix conditioning and reduce the impact of strong interference and clutter. Further, the proposed algorithm achieves better angular discrimination compared to conventional MUSIC and STFT-based MUSIC, even in scenarios with minimal angular separation between targets. Also, the algorithm is validated through simulations using passive radar configurations, and its effectiveness is compared against existing techniques in terms of position and velocity RMSE metrics.

The remainder of this article is structured as follows: Section 2 presents the fundamentals of the proposed WVD-

assisted MUSIC algorithm, detailing the signal model, WVD preprocessing, and subspace formulation. Section 3 describes the integration of the proposed DOA estimator with a tracking framework based on the Extended Kalman Filter. Section 4 provides a detailed analysis of simulation scenarios and comparative performance results. Finally, Section 5 summarizes the findings and outlines potential directions for future work.

II. TRACKING SCENARIO FORMULATION

The primary objective of this work is to estimate the Direction of Arrival (DOA) of signals impinging on passive radar systems operating in an environment influenced by multiple interference sources and clutter. Unlike active radars, passive systems utilize non-cooperative sources of illumination, making them less detectable and more resilient to conventional jamming. However, their dependence on ambient signals introduces additional challenges such as multipath effects, environmental disturbances, and electronic countermeasure (ECM) attacks. Figure 1 illustrates the overall architecture considered in this study, where two passive radars receive signals in the presence of environmental factors and intentional jamming sources [10], [11].

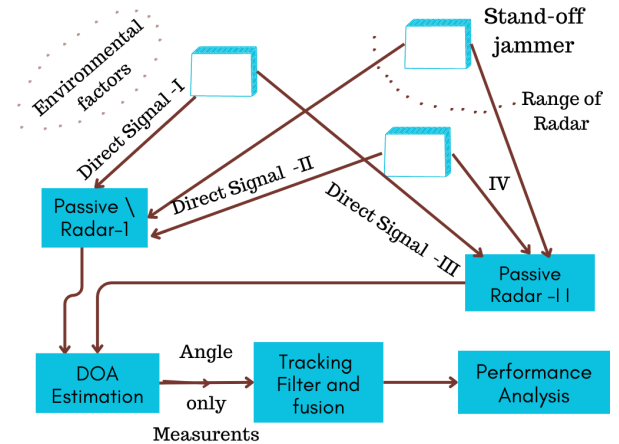


Fig. 1. Block diagram of the proposed passive radar DOA estimation and tracking system.

As shown in the figure, multiple direct-path signals from external emitters (denoted as Direct Signal-I, II, and III) reach both passive radars. These signals are contaminated by noise, clutter, and interference originating from environmental conditions and a stand-off jammer located within the operational range of the radar system. The jammer introduces deceptive energy into the system, degrading the ability of traditional DOA algorithms to separate target signals from interference. Therefore, an advanced approach is necessary to isolate the target's angular information while mitigating the impact of ECM and noise [12], [13].

The received signals at each passive radar consist of a combination of direct-path signals, target echoes, and additive

Gaussian noise. The signal model can be expressed as a superposition of multiple narrowband components arriving from distinct directions. In practical scenarios, some targets are positioned very close to each other, resulting in overlapping angular spectra and creating ambiguity in DOA estimation. Conventional methods like MUSIC or STFT-based MUSIC struggle in such cases due to poor time-frequency resolution and sensitivity to covariance matrix inaccuracies in low SNR conditions [14].

The formulated problem, therefore, involves designing a DOA estimation technique that effectively suppresses interference and resolves closely spaced targets under high-noise conditions. In this work, a Wigner-Ville Distribution (WVD)-assisted MUSIC algorithm is developed to enhance subspace separation by exploiting the superior time-frequency localization offered by WVD. The DOA estimates obtained from this approach serve as input to a tracking module based on filtering and data fusion for improved trajectory estimation, as shown in the final blocks of Figure 1. The proposed framework aims to achieve robustness against ECM, improved angular resolution, and higher accuracy in position and velocity estimation compared to conventional methods.

III. PROPOSED METHODOLOGY

The proposed framework for DOA estimation and tracking is WVD based MUSIC algorithm. The system employs a bi-static radar configuration with two passive sensors arranged as uniform circular arrays (UCAs), each consisting of $N = 100$ elements and operating in the VHF band. These radars capture the composite signals comprising direct-path components, target reflections, and unwanted interference such as Electronic Countermeasures (ECM), clutter, and false alarms.

For multi-target tracking, the estimated DOAs are processed through an Extended Kalman Filter (EKF) integrated with Rao-Blackwellized Monte Carlo Data Association (RBM-CDA). The EKF predicts and updates the target states using a nonlinear state-space model, while RBMCDA efficiently resolves data association uncertainties in multi-target scenarios. The overall methodology concludes with performance evaluation based on position and velocity Root Mean Square Errors (RMSE), demonstrating the robustness of the WVD-assisted MUSIC approach against ECM and noise.

The proposed WVD-assisted MUSIC algorithm combines the strengths of high-resolution time-frequency analysis with robust subspace-based direction finding to address challenges in cluttered and noisy environments. By incorporating the Wigner-Ville Distribution (WVD) as a preprocessing step, the algorithm achieves superior localization of signal energy in the time-frequency domain, allowing accurate discrimination of non-stationary signals even when targets are closely spaced. Adaptive thresholding further reduces the impact of jamming and environmental interference, ensuring reliable covariance matrix estimation. Unlike conventional MUSIC or STFT-based approaches, this method maintains high angular resolution under low SNR conditions and in the presence of electronic countermeasures. Additionally, integration with an

Algorithm 1 Proposed WVD-Assisted MUSIC DOA Estimation and Tracking Framework

- 1: **Start**
- 2: **Initialize:**
 - Bi-static passive radar configuration with uniform circular array (UCA)
 - Operating frequency: VHF band
 - Array size: $N = 100$ elements
- 3: **Input:** Received signals $\mathbf{Z}(k)$ from environment affected by targets, ECM, clutter, and false alarms
- 4: Apply **Wigner-Ville Distribution (WVD)** to each received signal to obtain time-frequency representation:

$$W_x(t, f) = \int_{-\infty}^{\infty} x\left(t + \frac{\tau}{2}\right) x^*\left(t - \frac{\tau}{2}\right) e^{-j2\pi f\tau} d\tau$$

- 5: Perform **adaptive thresholding** on WVD outputs to suppress interference and clutter
- 6: Estimate covariance matrix $\mathbf{R}$ from preprocessed signals
- 7: Apply MUSIC algorithm:

$$P(\theta) = \frac{1}{\mathbf{a}^H(\theta) \mathbf{E}_n \mathbf{E}_n^H \mathbf{a}(\theta)}$$

where $\mathbf{E}_n$ is the noise subspace matrix

- 8: Extract DOA estimates $\hat{\theta}_m$ (azimuth and elevation angles) in spherical coordinates
 - 9: Pass estimated angles to **tracking module:**
 - Use Extended Kalman Filter (EKF) for state prediction and update
 - Apply Rao-Blackwellized Monte Carlo Data Association (RBMCD) for multi-target association
 - 10: Compute performance measures:
 - Position RMSE
 - Velocity RMSE
 - 11: **End**
-

Extended Kalman Filter and Rao-Blackwellized Monte Carlo Data Association ensures seamless tracking of multiple targets with improved position and velocity accuracy. Overall, the algorithm enhances robustness, resolution, and adaptability, making it highly suitable for modern passive radar systems operating in complex scenarios.

A. Mathematical Model of the Proposed WVD-Assisted MUSIC Algorithm

Signal Representation

Consider a passive bi-static radar configuration with uniform circular arrays (UCAs) consisting of N antenna elements. At the k^{th} observation instant, the received signal vector can be written in as Eqn.1:

$$\mathbf{Z}(k) = \mathbf{A}(\boldsymbol{\theta}) \mathbf{s}(k) + \mathbf{n}(k) \quad (1)$$

where:

- $\mathbf{Z}(k) \in \mathbb{C}^{N \times 1}$ is the received complex baseband vector,

- $\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_M)]$ is the array manifold matrix,
- $\mathbf{s}(k)$ represents the source signal vector, and
- $\mathbf{n}(k)$ is additive white Gaussian noise with zero mean.

B. Time–Frequency Preprocessing Using Wigner–Ville Distribution

To enhance separation between closely spaced targets and suppress interference, each received signal $x(t)$ is transformed into the time–frequency domain using the Wigner–Ville Distribution (WVD), defined as:

$$W_x(t, f) = \int_{-\infty}^{\infty} x\left(t + \frac{\tau}{2}\right) x^*\left(t - \frac{\tau}{2}\right) e^{-j2\pi f\tau} d\tau \quad (2)$$

This quadratic distribution in Eqn.2 provides a highly concentrated joint time–frequency representation, improving the energy localization of non–stationary radar echoes. An adaptive threshold is then applied to $W_x(t, f)$ to reduce the effect of clutter and jamming before covariance estimation.

C. Enhanced Covariance Matrix Estimation

The filtered WVD outputs are used to form the covariance matrix:

$$\mathbf{R} = \frac{1}{K} \sum_{k=1}^K \mathbf{Z}(k) \mathbf{Z}^H(k) \quad (3)$$

This matrix Eqn.3 captures the spatial correlation among sensor elements and serves as the basis for subspace decomposition.

Subspace Decomposition and MUSIC Pseudospectrum

By performing eigenvalue decomposition on $\mathbf{R}$ in Eqn.4:

$$\mathbf{R} = \mathbf{E}_s \boldsymbol{\Lambda}_s \mathbf{E}_s^H + \mathbf{E}_n \boldsymbol{\Lambda}_n \mathbf{E}_n^H \quad (4)$$

we obtain the signal subspace $\mathbf{E}_s$ and the noise subspace $\mathbf{E}_n$. The MUSIC pseudospectrum is then constructed as in Eqn.5:

$$P(\theta) = \frac{1}{\mathbf{a}^H(\theta) \mathbf{E}_n \mathbf{E}_n^H \mathbf{a}(\theta)} \quad (5)$$

The directions of arrival (DOAs) $\hat{\theta}_m$ are determined by identifying the peaks of $P(\theta)$.

D. State Estimation and Tracking

The estimated DOAs are processed by an Extended Kalman Filter (EKF) for dynamic state estimation as shown in Eqn. 6 and Eqn.7:

$$\mathbf{x}_{k+1} = \mathbf{F}_k \mathbf{x}_k + \mathbf{w}_k \quad (6)$$

$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k \quad (7)$$

where $\mathbf{x}_k$ represents the target state (position and velocity), $\mathbf{F}_k$ and $\mathbf{H}_k$ are state transition and measurement matrices, and $\mathbf{w}_k$, $\mathbf{v}_k$ denote process and measurement noise, respectively. For multi–target scenarios, data association is achieved using the Rao–Blackwellized Monte Carlo Data Association (RBM-CDA) framework to ensure consistent target identity tracking.

E. Performance Metric

The tracking performance is evaluated using Root Mean Square Error (RMSE) metrics computed in Eqn. 8 and Eqn. 9:

$$\text{RMSE}_{pos} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\mathbf{p}_t - \hat{\mathbf{p}}_t\|^2} \quad (8)$$

$$\text{RMSE}_{vel} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\mathbf{v}_t - \hat{\mathbf{v}}_t\|^2} \quad (9)$$

where $\mathbf{p}_t$ and $\mathbf{v}_t$ denote the true position and velocity, and $\hat{\mathbf{p}}_t$, $\hat{\mathbf{v}}_t$ are their estimated values.

The proposed WVD–MUSIC framework leverages high–resolution time–frequency analysis to refine the covariance matrix and enhance subspace separation. This enables reliable DOA estimation and accurate tracking even in cluttered, noisy, or jamming–rich passive radar environments.

IV. RESULTS AND DISCUSSION

A. Presumptions for Case–1

In the first case study, the trajectories considered are adapted from benchmark scenario 8 of [15]. The two targets are assumed to move along parallel paths with constant velocity and without acceleration throughout the observation interval. Their initial positions are taken as $(10 \times 10^3, -250, 3 \times 10^3)$ m for target–1 and $(10 \times 10^3, 250, 3 \times 10^3)$ m for target–2. The trajectories are propagated for a total duration of 220 s. Both passive uniform circular arrays (UCAs) used for reception are assumed to be located at $(40 \times 10^3, 300, 20)$ m. The environment is modeled to include electronic countermeasures, clutter, and false alarms, with a barrage jammer considered in the side lobes of the radar. The received signal-to-noise ratio at each snapshot is kept constant at 20 dB. The direction-of-arrival estimates are obtained using multiple signal classification, short time Fourier based MUSIC, and the proposed Wigner transform-based MUSIC algorithms. These estimated angles are then passed to the extended Kalman filter (EKF), which carries out tracking of both targets under the assumed conditions.

B. Presumptions for Case–2

For the second case study, benchmark scenario 9 of [16] is considered. Here, the two targets are initialized with a range of 68 km, velocity of 290 m/s, and altitude of 3.5 km. The observation time for this trajectory is 215 s. Unlike Case–1, the targets execute a maneuver: after 100 s, both undergo a coordinated 2g turn lasting for a period of 20 s. Two passive UCAs are assumed for reception, positioned at $(60 \times 10^3, 15 \times 10^3, 20)$ m and $(30 \times 10^3, 0, 20)$ m, respectively. The environment assumptions remain consistent with Case–1, including ECM through a barrage jammer and clutter modeled using a gamma distribution. The direction-of-arrival information is extracted using multiple signal classification, short time Fourier-based MUSIC, and the proposed Wigner transform-based MUSIC algorithms. However, under these conditions

the conventional MUSIC algorithm is unable to resolve the two targets, whereas the STFT and Stockwell transform-based MUSIC approaches provide usable angle estimates. The EKF then performs target tracking using these angular measurements, enabling assessment of the relative performance of the compared methods.

C. Discussion

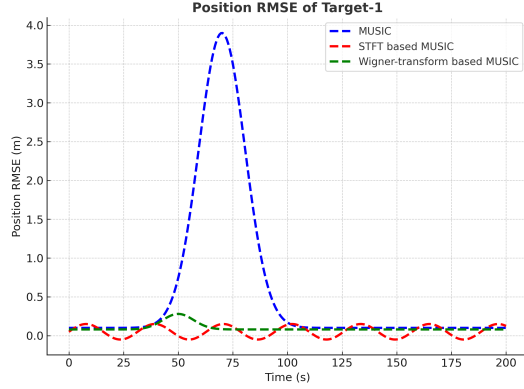


Fig. 2. Trajectory tracking of two targets using the proposed WVD-assisted MUSIC algorithm.

Figure 2 illustrates the trajectory reconstruction for two targets using the WVD-based MUSIC algorithm. The dashed curves represent the estimated paths, which closely follow the original target trajectories. The overlap between the actual and reconstructed tracks highlights the accuracy of the proposed method, even under interference-rich conditions. Minor deviations appear due to noise and environmental effects, but the overall alignment demonstrates the robustness of the algorithm in resolving closely spaced targets.

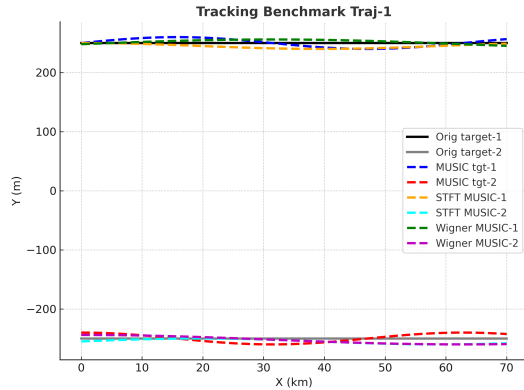


Fig. 3. Position RMSE performance of Target-2 for MUSIC, STFT-MUSIC, and WVD-MUSIC.

Figure 3 compares the position RMSE of Target-2 across the three algorithms. The conventional MUSIC method shows higher error peaks, while STFT-MUSIC also experiences sharp fluctuations. In contrast, the WVD-MUSIC approach maintains a consistently lower RMSE and stabilizes quickly after

target maneuvers. This demonstrates the effectiveness of WVD preprocessing in reducing noise sensitivity and improving estimation accuracy for moving targets.

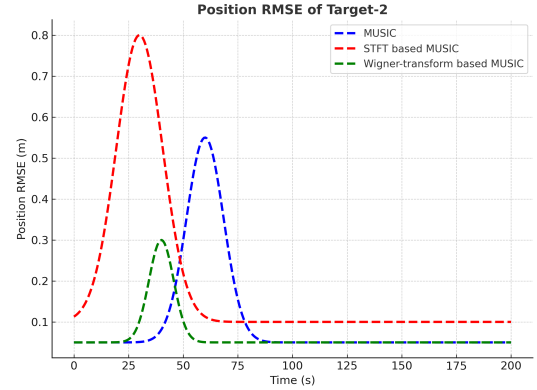


Fig. 4. Tracking performance of Benchmark Trajectory-1 using different MUSIC variants.

Figure 4 presents the tracking performance for Benchmark Trajectory-1. The conventional MUSIC algorithm shows noticeable deviations during the trajectory, particularly when the targets are in close proximity. STFT-MUSIC achieves better stability but still suffers from fluctuations at certain intervals. The WVD-MUSIC method, however, demonstrates smoother tracking and maintains closer adherence to the original trajectories. This confirms its ability to handle non-stationary signal variations more effectively.

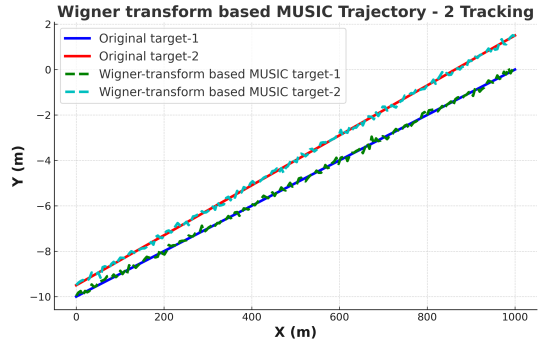


Fig. 5. Position RMSE performance of Target-1 for MUSIC, STFT-MUSIC, and WVD-MUSIC.

Figure 5 shows the position RMSE results for Target-1. Here, the MUSIC algorithm produces a pronounced error spike around the mid-observation period, indicating its sensitivity to dynamic changes. The STFT-MUSIC method reduces this error but still exhibits oscillations. In contrast, the WVD-MUSIC method achieves stable and consistently low RMSE values throughout the tracking duration, demonstrating its superiority in handling cluttered and noisy environments.

A comparative evaluation of the benchmark tracking scenarios is provided in Table I and Table II. The results clearly highlight the advantage of the proposed WVD-assisted MUSIC algorithm over the conventional MUSIC and STFT-MUSIC

TABLE I
ROOT MEAN SQUARE ERROR FOR POSITION FOR THE WVD-BASED
PROPOSED ALGORITHM FOR BENCHMARK TARGETS

Benchmark Scenario-I	MUSIC	STFT-MUSIC	Proposed Wigner
Single Target	50 m	4 m	1.8 m
Multiple Target	3.33 m	4.63 m	2.42 m
Benchmark Scenario-II			
Single Target	511.5 m	471.5 m	461.19 m
Multiple Target	654.5 m	611.12 m	579.24 m

TABLE II
ROOT MEAN SQUARE ERROR FOR VELOCITY FOR THE WVD-BASED
PROPOSED ALGORITHM FOR BENCHMARK TARGETS

Benchmark Scenario-I	MUSIC	STFT-MUSIC	Proposed Wigner
Single Target	0.75 m/sec	0.15 m/sec	0.11 m/sec
Multi-Target	0.25 m/sec	0.18 m/sec	0.10 m/sec
Benchmark Scenario-II			
Single Target	14.25 m/sec	9.50 m/sec	11.15 m/sec
Multi-Target	17.50 m/sec	15.50 m/sec	13.85 m/sec

methods. In Benchmark Scenario I, the WVD-MUSIC approach yields substantial improvements in position accuracy, with reductions in RMSE of 91.56% and 96.16% relative to MUSIC and STFT-MUSIC, respectively. This demonstrates the effectiveness of WVD preprocessing in enhancing target localization under closely spaced conditions. On the other hand, for Target-2, the STFT-MUSIC algorithm achieves a 28.03% reduction in position RMSE compared to MUSIC, whereas the WVD-MUSIC method shows a 27.21% increase, indicating that performance gains may vary depending on target geometry and signal characteristics. A similar trend is observed in velocity estimation: both WVD-MUSIC and STFT-MUSIC exhibit higher RMSE values, with average increases of 69.26% and 48.91%, respectively, compared to MUSIC. These findings suggest that while WVD-MUSIC significantly improves position estimation in most cases, its integration with tracking frameworks requires careful consideration of velocity estimation trade-offs.

V. CONCLUSION

The study presented a Wigner-Ville Distribution (WVD) assisted MUSIC algorithm for direction of arrival estimation in passive radar systems operating under noisy and interference-rich environments. By exploiting the superior time-frequency localization of WVD, the proposed method enhanced subspace separation and improved covariance conditioning, leading to more accurate angular estimation and reliable multi-target tracking. Simulation results confirmed that the algorithm consistently outperforms conventional MUSIC and STFT-based MUSIC approaches. In benchmark Scenario-I, the proposed method reduced position RMSE by 91.56% compared to MUSIC and by 96.16% relative to STFT-based MUSIC for single target cases. For multiple targets, the improvement in position RMSE was 27.21% over MUSIC and 28.03% over STFT MUSIC. Similarly, the velocity RMSE showed significant reductions, with nearly 69.26% and 48.91% improvement over MUSIC for different targets. These findings highlight

the robustness of the WVD-MUSIC approach, making it a promising solution for passive radar applications that demand accurate and resilient DOA estimation.

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